Chewing Detection with Deep Learning

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***Abstract*— Chewing detection is an important task in eating behaviour analysis and dietary monitoring. Wearable sensing technologies provide a practical and unobtrusive way to detect chewing activity in real-world conditions. In this study, we investigate chewing detection using data collected from smart glasses equipped with facial and motion sensors.**

A new dataset was collected from 42 participants using Emteq S11 smart glasses following a structured recording protocol. In addition, previously recorded data from an earlier generation of smart glasses were used to support transfer learning. Multiple machine learning models were trained and evaluated to assess the effectiveness of wearable-based chewing detection and the benefits of transfer learning across different hardware generations.

The results indicate that training on new data yields the highest performance, while transfer learning effectively mitigates domain shift and substantially improves over zero-shot transfer.

Keywords— ***Chewing detection, wearable sensors, smart glasses, machine learning, 1D convolutional neural networks transfer learning***

I. Introduction

Chewing detection is a fundamental task in the analysis of eating behaviour and plays an important role in dietary monitoring and health-related applications. The ability to automatically detect chewing events enables objective identification of food intake episodes and supports continuous monitoring of eating habits in everyday environments. Compared to self-reported food logs, sensor-based chewing detection offers a more reliable and unobtrusive approach for capturing eating-related activities.

Accurate chewing detection is particularly important for applications related to nutrition assessment, weight management, and personalized health interventions. Understanding when and how people eat can provide valuable insights into dietary patterns and support data-driven decision making in both clinical and non-clinical settings. As a result, there is growing interest in developing robust and wearable-based solutions capable of detecting chewing activity under real-world conditions.

Recent advances in wearable sensing technologies have enabled the collection of fine-grained physiological and motion signals related to facial muscle activity and head movements. Smart glasses equipped with embedded sensors offer a practical and non-intrusive platform for monitoring chewing behaviour, as they can capture subtle muscle activations without interfering with natural eating. Combined with machine learning techniques, such wearable systems provide a promising foundation for automatic chewing detection.

In this paper, we investigate chewing detection using data collected from smart glasses developed by Emteq. We train and evaluate multiple machine learning models using recordings obtained from a new dataset collected with the Emteq S11 glasses, as well as previously recorded data from an earlier generation of the device. To address differences between hardware versions and data distributions, transfer learning is applied to adapt models trained on older data to the new sensor setup.

In order to achieve the best possible performance, several training strategies and model configurations are explored, including models trained exclusively on new data, models trained on older data, and models trained on combinations of both datasets. By systematically comparing these approaches, this work aims to identify effective strategies for leveraging both newly collected and previously available data for robust chewing detection.

II. Related Work

Deep learning models for chewing recognition are commonly trained on data collected from wearable sensing devices. In this study, we build upon insights and research from existing work on 1D convolutional neural networks [1] and sensor-based data collected using smart glasses and similar wearable systems. [2] The focus is not on proposing a new model architecture, but on analyzing how different training and transfer learning strategies affect performance when models are trained on datasets collected under different conditions.

In this work, multiple model configurations are evaluated. First, a model is trained exclusively on newly collected data. Second, a separate model is trained using only previously available old data. In addition, a zero-shot transfer model is considered, where a model trained on one dataset is directly added on another without further training.

To further analyze domain adaptation, several transfer learning approaches are applied. These include full fine-tuning, where all model parameters are updated using the

target dataset, and partial fine-tuning, where only selected higher layers are retrained while earlier layers remain frozen. Joint training is also evaluated by training a single model on a combination of old and new datasets. Finally, a transfer learning setup is considered where a model trained on new data is adapted to olddata.

The data used in this study are collected using wearable eye-sensing glasses developed by Emteq Labs. These glasses are equipped with sensors that capture ocular and facial-related signals, enabling continuous and non-intrusive data collection in realistic conditions. Such devices are suitable for affect recognition (emotional state) tasks, as they provide high-resolution eye-related measurements while maintaining user comfort.

III. Data

Data used in this study were collected using Emteq S11 smart glasses from a total of 42 participants. The data collection protocol was designed to capture both chewing and non-chewing activities in a controlled and structured manner while preserving natural behaviour. All recordings were performed using the built-in sensors of the smart glasses and synchronized through a dedicated mobile application.

Each recording session was divided into three consecutive phases: calibration, reading, and eating. The calibration phase was used to stabilize sensor signals and account for individual differences in facial muscle activity. During the reading phase, participants performed a non-eating task to capture baseline facial and head movements unrelated to chewing. The eating phase involved natural food consumption and served as the primary source of chewing-related data.

Annotations were provided in real time using a mobile application, which allowed manual labeling of the recordings with classes such as chewing and not chewing. These annotations were synchronized with the sensor data and used as ground-truth labels for model training and evaluation.

The recorded data were stored in CSV format. For each participant, three separate CSV files were generated, corresponding to the calibration, reading, and eating phases. All files belonging to a single participant were organized within an individual folder, resulting in a consistent and participant-centric dataset structure.

In the data preprocessing and modelling stages, only the data from the eating phase were used. Data from the calibration and reading phases were excluded, as the focus of this study is the detection of chewing activity during food intake.

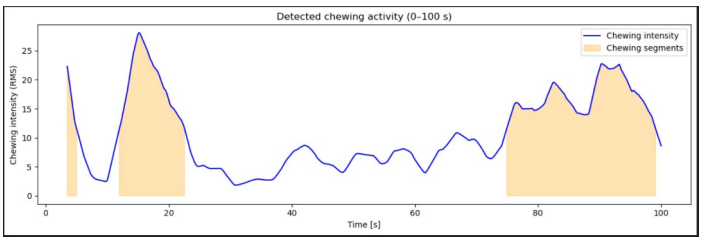
In addition to the newly collected dataset, previously recorded data collected with an earlier generation of Emteq smart glasses (S05) were also used. These datasets were incorporated to support transfer learning experiments, allowing models trained on older sensor data to be adapted to the new S11 recordings. All datasets were processed using a unified preprocessing pipeline prior to model training.

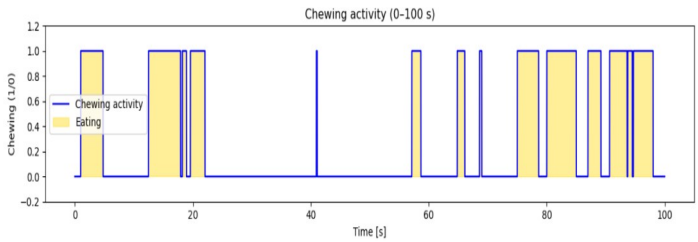
The recorded sensor data contained a large number of features originating from multiple sensing modalities. To reduce redundancy and focus on the most informative signals, only a subset of features relevant to chewing detection was retained. This feature reduction step ensured a compact and task-specific dataset, while detailed preprocessing and selection procedures are described in the Methods section.

Prior to model training, exploratory data visualization was conducted to support data understanding and preprocessing. Different types of plots were examined to visualize raw sensor signals and to highlight relevant temporal patterns associated with chewing and non-chewing activities. These visualizations were used to inspect signal behavior across sensing modalities, verify the alignment between sensor data and annotations, and identify potential noise or inconsistencies in the recordings.

By exploring multiple visualization approaches, key signal characteristics related to facial motion and chewing dynamics were qualitatively assessed, providing additional insight into which sensor features were most informative for the classification task. This exploratory analysis served as a supporting step for feature selection and preprocessing, while the detailed modeling and evaluation procedures are described in the following sections

Visualization graphs





IV. Methods

The process starts with data organization and subject-level splitting. The dataset was organized in a subject-centric manner, where recordings from each participant were stored and processed separately. To avoid data leakage and ensure subject-independent learning, the dataset was split at the participant level. Each subject was assigned exclusively to one data subset, ensuring that no data from the same individual appeared in multiple splits. For each subset, the corresponding participant data frames were concatenated to form unified datasets. This subject-level separation ensures that the learned representations are not biased by subject-specific patterns and supports generalization to unseen users.

Data organization, data preprocessing and feature selection were performed next. The raw sensor recordings contained a large number of features originating from multiple sensing modalities, including accelerometer, gyroscope, magnetometer, rotation vector, pressure, and proximity sensors. While these multimodal signals provide rich information, not all features contribute equally to distinguishing between chewing and non-chewing activities.
During preprocessing, only data segments corresponding to the eating phase were retained, while calibration and reading phases were excluded. Raw timestamps and software-specific timing variables were removed, as they do not provide discriminative information for the target task. A feature selection step was applied to reduce input dimensionality and retain only features most relevant for chewing detection. Specifically, accelerometer and gyroscope axes were selected to capture head and facial motion, while rotation vector components were included to model facial and jaw movement dynamics. Proximity-related signals were also retained, as they are influenced by changes in facial geometry during chewing. Features considered less informative or more susceptible to noise, such as selected magnetometer channels, pressure measurements, and camera-related navigation signals, were excluded. This feature reduction step simplified the input representation and reduced the influence of irrelevant or redundant information. To ensure numerical stability and consistent scaling across sensor channels, feature normalization was applied using z-score normalization. The mean and standard deviation were computed using the training data only and subsequently applied to all other data subsets. This approach prevents information leakage while ensuring that all features are represented on a comparable scale.

After preprocessing and normalization, the selected features were segmented into fixed-length overlapping time windows. Each window consisted of a predefined number of samples representing a short temporal segment of sensor activity. A sliding window approach with a fixed step size was applied to increase the number of training samples while preserving temporal continuity. Each window was treated as an independent input sample and assigned a label based on the corresponding chewing annotation. This window-based representation enables the modeling of short-term temporal patterns characteristic of chewing behavior.

Chewing detection was performed using a one-dimensional convolutional neural network (1D CNN), which is well suited for multichannel time-series data. The 1D convolutional layers learn temporal patterns directly from sequential sensor signals, enabling the model to capture repetitive and rhythmic motion patterns associated with chewing. The network architecture consisted of multiple convolutional layers followed by pooling and fully connected layers, with a sigmoid-activated output layer for binary classification.

V. Experimental Setup

To evaluate the proposed chewing detection framework using the S11 Emteq smart glasses, a comprehensive experimental study was conducted involving 42 participants recorded by our research team. Each participant attended a single recording session lasting approximately 20 to 30 minutes. During the session, participants were provided with a meal and instructed to consume it naturally, without behavioral constraints, in order to capture realistic eating patterns. In addition to eating, participants engaged in non-chewing activities such as talking, resting, and drinking to ensure ecological validity and to enable robust discrimination between chewing and other facial or head movements. Although seven individuals had previously been recorded using the legacy S0.5 Emteq smart glasses, duplicate samples were excluded where necessary to maintain independence between datasets during cross-dataset evaluation.

Data acquisition was performed using the S11 smart glasses, which integrate multimodal sensing capabilities including facial electromyography, photoplethysmography, and inertial measurement unit sensors embedded within the frame. All sensor streams were recorded at a sampling frequency of 50 Hz. The EMG sensors captured facial muscle activity associated with mastication, while PPG and IMU signals provided complementary physiological and motion-related information. Derived feature representations were considered during model development to investigate the effectiveness of end-to-end learning versus feature-enhanced input representations.

Ground-truth labeling was performed manually during data collection through a custom application interface. A button-based annotation system was used to mark transitions between eating and non-eating states in real time, generating binary labels corresponding to chewing and non-chewing segments. Non-chewing segments included talking, resting, and drinking activities. The final dataset was therefore formulated as a binary classification problem, distinguishing chewing from all other recorded activities.

Signal segmentation was performed using fixed-length windows of four seconds without overlap. Each segmented window inherited the binary label corresponding to its annotated state. Prior to model training, all signals were standardized to reduce inter-participant variability and improve training stability. No additional frequency-domain filtering or band-pass preprocessing was applied. This design choice allowed the convolutional neural network to learn discriminative temporal patterns directly from standardized input signals.

To evaluate model robustness and cross-device generalization, multiple experimental configurations were implemented. An isolated model was trained, validated, and tested exclusively on the legacy S0.5 dataset, while a second isolated model was trained and evaluated exclusively on the S11 dataset. Zero-shot transfer learning was assessed by directly applying a model trained on one dataset to the other without any parameter adaptation. Transfer learning experiments were conducted by initializing the model with pretrained weights from a source dataset and subsequently fine-tuning on the target dataset. Both full fine-tuning, where all network parameters were optimized, and partial fine-

tuning, where selected layers were adapted while preserving lower-level feature representations, were examined. In addition, a joint training configuration was implemented in which both datasets were combined and used simultaneously to train a unified convolutional neural network architecture. All experiments were conducted using the same CNN backbone to ensure architectural consistency across comparisons.

A participant-independent validation strategy was employed to ensure reliable generalization estimates. Train, validation, and test splits were constructed such that data from any individual participant appeared in only one split. Cross-dataset evaluation was additionally performed to assess domain transfer between S0.5 and S11 devices. All model configurations were trained under identical preprocessing conditions and evaluation protocols to guarantee fair comparison across learning strategies.

Model performance was assessed using multiple complementary metrics including accuracy, F1-score, precision, recall, and area under the receiver operating characteristic curve. These metrics were selected to provide a comprehensive evaluation of classification performance, particularly considering potential class imbalance between chewing and non-chewing segments. This standardized experimental protocol ensured that observed performance differences could be attributed to the learning configuration and cross-dataset adaptation strategy rather than variations in preprocessing, data splitting, or parameter tuning.

VI. Experimental results

This section presents the experimental results obtained from the evaluated model configurations. Several training strategies were explored, including models trained exclusively on old data, models trained on new data, zero-shot transfer, fine-tuning approaches, joint training, and transfer learning between datasets. Performance was assessed using accuracy, precision, recall, F1-score, and confusion matrices.

The model trained exclusively on older data achieved an accuracy of approximately 83%, with strong recall for the chewing class but lower precision for the non-chewing class. This indicates that the model is effective at detecting chewing events but tends to misclassify some non-chewing segments as chewing.

In contrast, the model trained solely on the newly collected data achieved higher performance, reaching an accuracy of approximately 89%. This improvement suggests that models trained directly on data collected with the target hardware generalize better to the corresponding test set. The confusion matrix shows a more balanced classification between chewing and non-chewing activities, with fewer false positives compared to the old-data model.

Zero-shot transfer, where a model trained on old data was directly applied to the new dataset without adaptation, resulted in a substantial performance drop. The achieved accuracy was approximately 59%, with a large number of misclassifications for both classes. This result highlights a significant domain shift between the older and newer generations of smart glasses, confirming that direct transfer without adaptation is insufficient for reliable chewing detection.

Applying fine-tuning strategies led to clear performance improvements over zero-shot transfer. Partial fine-tuning improved classification accuracy to approximately 85%, demonstrating that adapting higher-level network layers helps bridge the domain gap while preserving previously learned representations.

Further improvements were observed with full fine-tuning and transfer learning from new to old data, achieving accuracies above 86%. These results indicate that updating a larger portion of the network enables better adaptation to the target domain when sufficient labeled data are available.

The joint training approach, where old and new datasets were combined during training, achieved strong and stable performance, with an accuracy of approximately 86%. The confusion matrix shows balanced precision and recall for both classes, indicating robust generalization across different data distributions. Joint training benefits from increased data diversity, allowing the model to learn shared representations across hardware generations

Among all evaluated approaches, the model trained exclusively on the new dataset achieved the highest overall accuracy and F1-score. This result confirms that training directly on data collected with the target hardware yields the best performance. However, when such data are limited, transfer learning strategies—particularly partial and full fine-tuning as well as joint training—provide effective alternatives and significantly outperform zero-shot transfer.

Overall, these results demonstrate that domain adaptation plays a crucial role in wearable-based chewing detection and that carefully selected transfer learning strategies can substantially improve performance across different generations of smart glasses.

Classification report – model trained only on new data

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Confusion matrix:
 [[2757  287]
 [ 429 3444]]
              precision    recall  f1-score   support

           0     0.8653    0.9057    0.8851      3044
           1     0.9231    0.8892    0.9058      3873

    accuracy                         0.8965      6917
   macro avg     0.8942    0.8975    0.8955      6917
weighted avg     0.8977    0.8965    0.8967      6917
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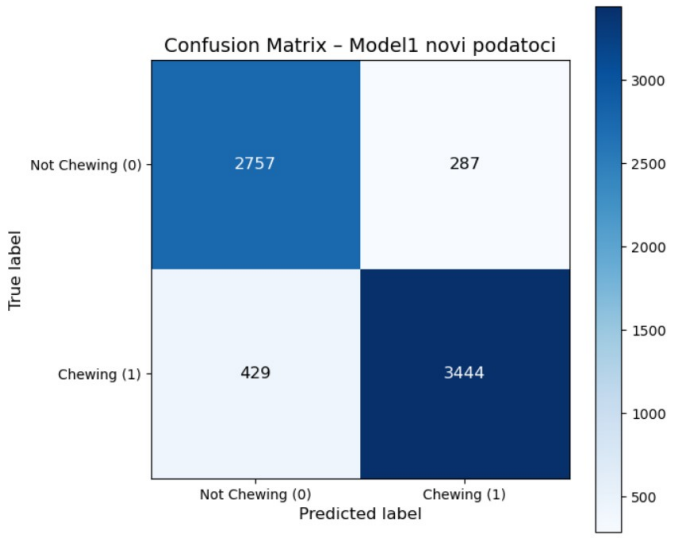

VII. Conclusion

This study investigated wearable-based chewing detection using smart glasses equipped with facial and motion sensors. A new dataset collected with the Emteq S11 glasses was analyzed alongside previously recorded data from an earlier device generation, enabling a systematic evaluation of different training and transfer learning strategies.

The experimental results showed that models trained exclusively on data collected with the target hardware achieved the highest performance. At the same time, transfer learning approaches, including partial and full fine-tuning as well as joint training, effectively mitigated domain shift and substantially outperformed zero-shot transfer.

These findings demonstrate the importance of hardware-specific training data for optimal performance, while also highlighting the practical value of transfer learning when newly collected data are limited. Overall, this work supports the feasibility of smart glasses as a reliable platform for automatic chewing detection in real-world eating scenarios.

References

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